

Lab 1A

Joseph Leone

2/27/26

CDA 3203 Computer Logic Design
Semester Spring 2026

Dr. Maria Petrie
Florida Atlantic University

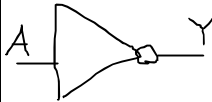
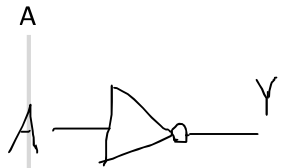
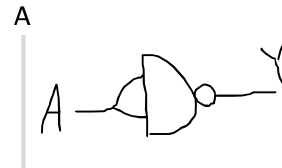
HANDWRITTEN SECTION: ALL the work in this part must be **HAND-WRITTEN/HAND-DRAWN.**

Design 8 circuits by completing below: **HAND-DRAW** the symbol for the gate, its Truth Table, the Canonical Sum of Products ($Y =$), the NOT-AND-OR Equivalent Circuit of its **Canonical Sum of Products** and its all-NAND Equivalent Circuit of its **Canonical Sum of Products** using 2-input or 3-input NANDS.

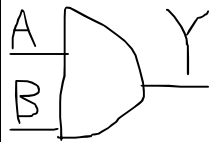
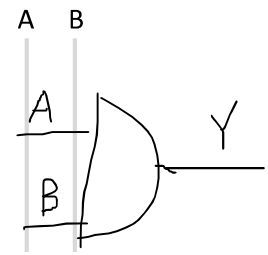
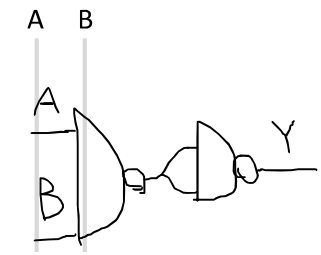
Images that are copied and pasted will not be accepted. To earn credit, you must ensure ALL circuits are correct and fully legible.

Images and text that are not HAND DRAWN/HANDWRITTEN will NOT be accepted.

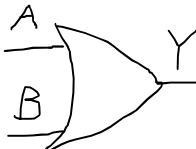
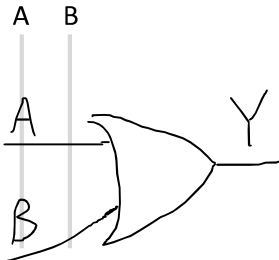
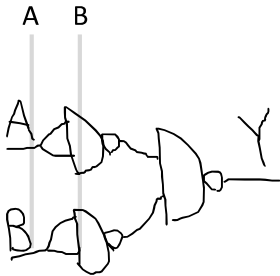
1.1 NOT gate. (5pts)

Draw NOT gate <i>(1pts)</i>	Truth Table and CANONICAL Sum of Products Equation <i>(2pts)</i>	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>						
Y= 	<table border="1"><thead><tr><th>A</th><th>NOT(A)</th></tr></thead><tbody><tr><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td></tr></tbody></table> $Y = \overline{A}$	A	NOT(A)	1	0	0	1		
A	NOT(A)								
1	0								
0	1								

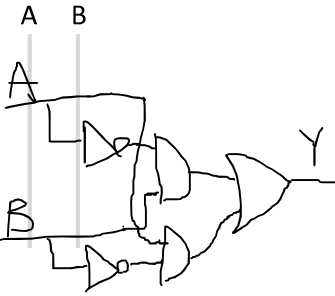
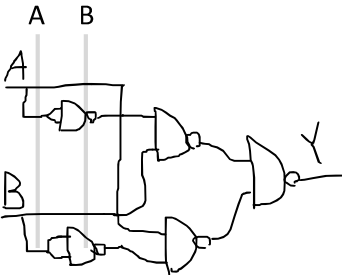
1.2- AND gate(5pts)

Draw AND gate <i>(1pts)</i>	Truth Table and CANONICAL Sum of Products Equation <i>(2pts)</i>	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>															
Y= 	<table border="1"><thead><tr><th>A</th><th>B</th><th>AND(A,B)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></tbody></table> $Y=AB$	A	B	AND(A,B)	0	0	0	0	1	0	1	0	0	1	1	1		
A	B	AND(A,B)																
0	0	0																
0	1	0																
1	0	0																
1	1	1																

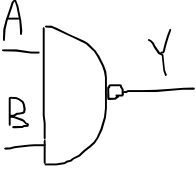
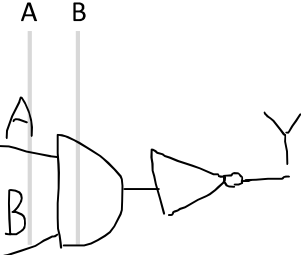
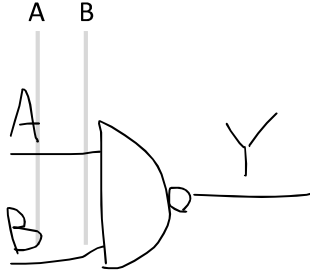
1.3- OR gate(5pts)

Draw OR gate (1pts)	Truth Table and CANONICAL Sum of Products Equation (2pts)	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation (1pts)	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation (1pts)															
Y= 	<table border="1"><thead><tr><th>A</th><th>B</th><th>OR(A,B)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></tbody></table> $Y = \bar{A}B + A\bar{B} + AB$	A	B	OR(A,B)	0	0	0	0	1	1	1	0	1	1	1	1		
A	B	OR(A,B)																
0	0	0																
0	1	1																
1	0	1																
1	1	1																

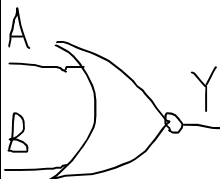
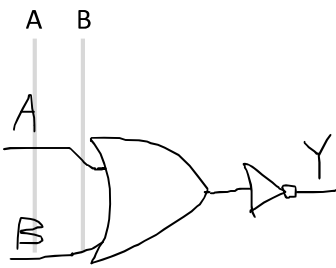
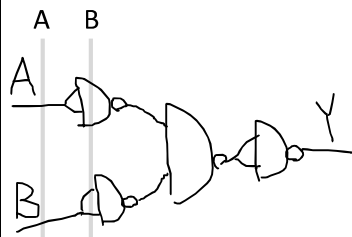
1.4- XOR gate(5pts)

Draw XOR gate <i>(1pts)</i>	Truth Table and CANONICAL Sum of Products Equation <i>(2pts)</i>	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>															
Y= 	<table border="1"><thead><tr><th>A</th><th>B</th><th>XOR(A,B)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></tbody></table> $Y = \bar{A}B + A\bar{B}$	A	B	XOR(A,B)	0	0	0	0	1	1	1	0	1	1	1	0		
A	B	XOR(A,B)																
0	0	0																
0	1	1																
1	0	1																
1	1	0																

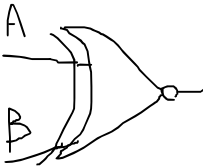
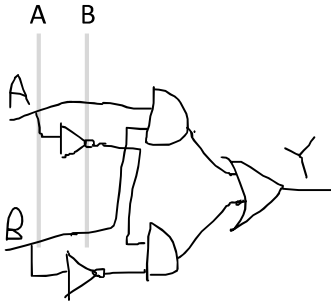
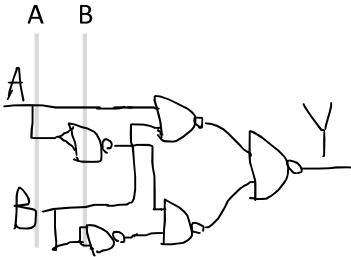
1.5- NAND gate(5pts)

Draw NANDgate (1pts)	Truth Table and CANONICAL of Products Equation (2pts)	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation (1pts)	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation (1pts)															
Y= 	<table border="1"><thead><tr><th>A</th><th>B</th><th>NAND(A,B)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></tbody></table> $Y = \bar{A}\bar{B} + \bar{A}B + A\bar{B}$	A	B	NAND(A,B)	0	0	1	0	1	1	1	0	1	1	1	0		
A	B	NAND(A,B)																
0	0	1																
0	1	1																
1	0	1																
1	1	0																

1.6- NOR gate(5pts)

Draw NOR gate (1pts)	Truth Table and CANONICAL Sum of Products Equation (2pts)	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation (1pts)	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation (1pts)															
Y= 	<table><thead><tr><th>A</th><th>B</th><th>NOR(A,B)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></tbody></table> $Y = \overline{A} \overline{B}$	A	B	NOR(A,B)	0	0	1	0	1	0	1	0	0	1	1	0		
A	B	NOR(A,B)																
0	0	1																
0	1	0																
1	0	0																
1	1	0																

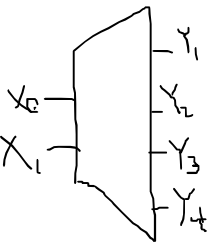
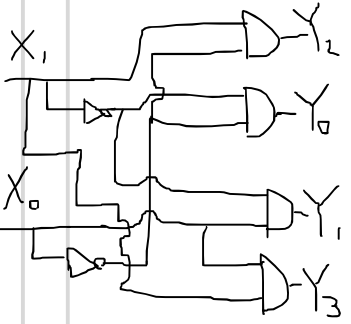
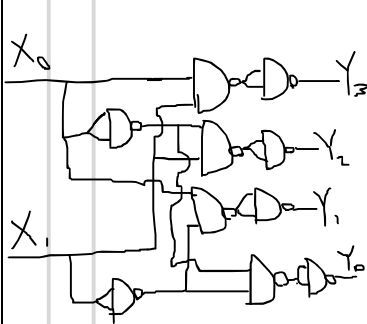
1.7- XNOR gate(5pts)

Draw XNOR gate gate <i>(1pts)</i>	Truth Table and CANONICAL Sum of Products Equation <i>(2pts)</i>	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>															
Y= 	<table border="1"><thead><tr><th>A</th><th>B</th><th>XNOR(A,B)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></tbody></table> $Y = \bar{A}\bar{B} + AB$	A	B	XNOR(A,B)	0	0	1	0	1	0	1	0	0	1	1	1		
A	B	XNOR(A,B)																
0	0	1																
0	1	0																
1	0	0																
1	1	1																

1.8- 2 to 1 Encoder or Multiplexer (Mux)(5pts)

Draw 2to1 Mux <i>(1pts)</i>	Truth Table and CANONICAL Sum of Products Equation <i>(2pts)</i>	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation <i>(1pts)</i>																																										
	<table border="1"><thead><tr><th>sel</th><th>X₁</th><th>X₀</th><th>Mux</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></tbody></table> OR <table border="1"><thead><tr><th>sel</th><th>Mux</th></tr></thead><tbody><tr><td>0</td><td>X₀</td></tr><tr><td>1</td><td>X₁</td></tr></tbody></table> $Y = \bar{S}X_0 + SX_1$	sel	X ₁	X ₀	Mux	0	0	0	0	0	0	1	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	1	1	1	1	1	sel	Mux	0	X ₀	1	X ₁		
sel	X ₁	X ₀	Mux																																										
0	0	0	0																																										
0	0	1	1																																										
0	1	0	0																																										
0	1	1	1																																										
1	0	0	0																																										
1	0	1	0																																										
1	1	0	1																																										
1	1	1	1																																										
sel	Mux																																												
0	X ₀																																												
1	X ₁																																												

1.9 -- 2 to 4 decoder or Demultiplexer (DMux)(5pts)

Draw 2to4 DMux	Truth Table and CANONICAL Sum of Products Equation	NOT-AND-OR Equivalent Circuit of the CANONICAL Sum of Products Equation	ALL-NAND Equivalent Circuit of the CANONICAL Sum of Products Equation																														
(1pts)	(2pts)	(1pts)	(1pts)																														
 Y3= Y2= Y1= Y0=	<table border="1"><thead><tr><th>X₁</th><th>X₀</th><th>Y₃</th><th>Y₂</th><th>Y₁</th><th>Y₀</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></tbody></table> $Y_3 = \overline{X_1} \cdot \overline{X_0}$$Y_1 = \overline{X_1} \cdot X_0$$Y_2 = X_1 \cdot \overline{X_0}$$Y_3 = X_1 \cdot X_0$	X ₁	X ₀	Y ₃	Y ₂	Y ₁	Y ₀	0	0	0	0	0	1	0	1	0	0	1	0	1	0	0	1	0	0	1	1	1	0	0	0		
X ₁	X ₀	Y ₃	Y ₂	Y ₁	Y ₀																												
0	0	0	0	0	1																												
0	1	0	0	1	0																												
1	0	0	1	0	0																												
1	1	1	0	0	0																												

QUARTUS SECTION: For the rest of Lab 1A, you will use Altera Quartus v9.1 to code and simulate multiple components in VHDL below. Please make sure all screenshots of code and simulations are LEGIBLE and not blurry!

YOU MUST MAKE SURE THAT YOUR NAME IS EMBEDDED ON TAB IN THE SCREENSHOTS!

Part 2.1(8pts)-Create a new project in Altera Quartus in your working directory **Lab1A_YourName** using VHDL to define a 3-input NAND circuit called **NAND3_YourName**, using all-NAND gates (**only 2-input NAND gates**).

Project Settings(1pts)

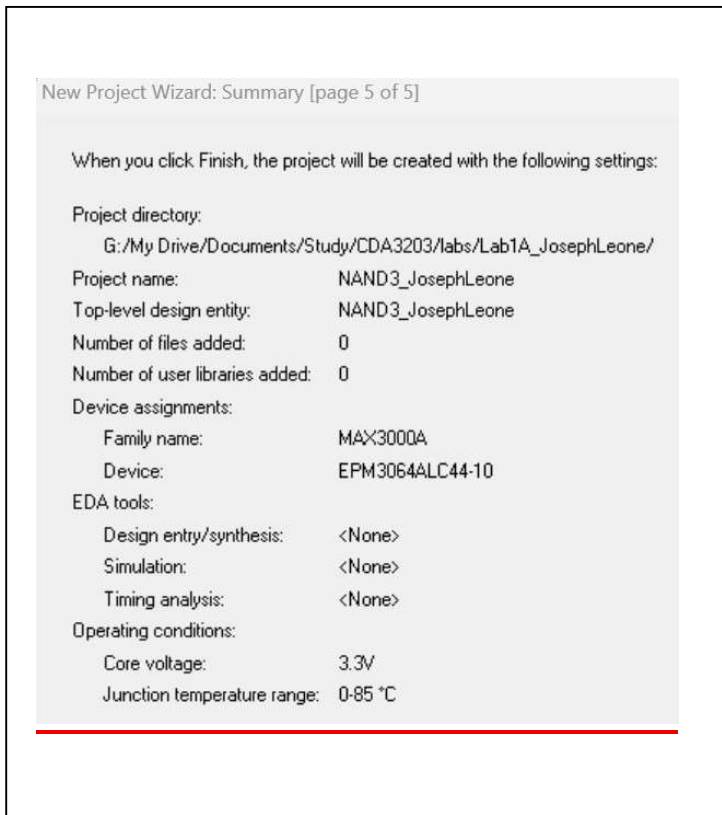
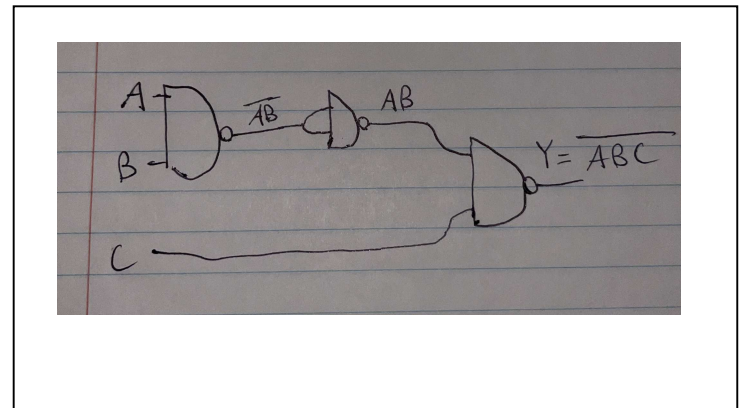


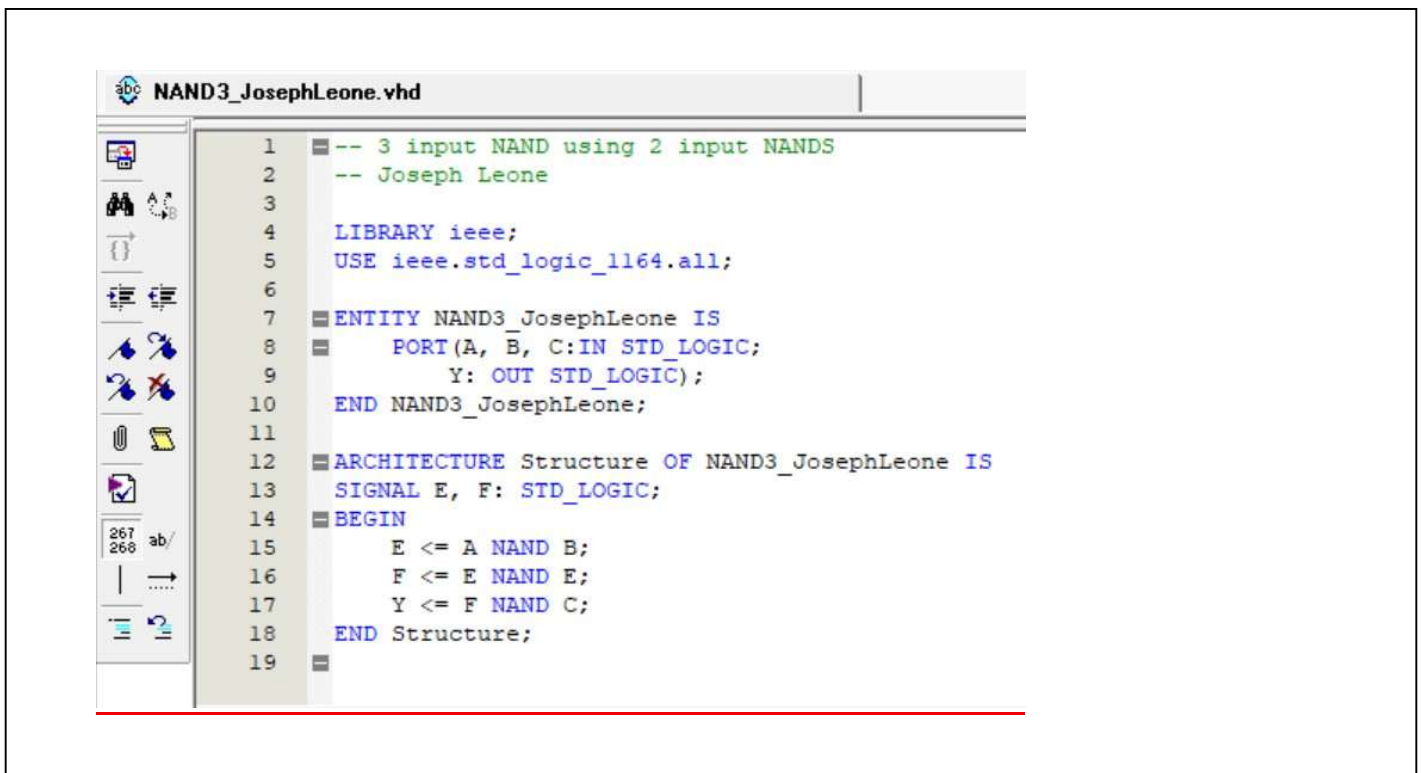
Diagram (HANDDRAWN)(1pts)



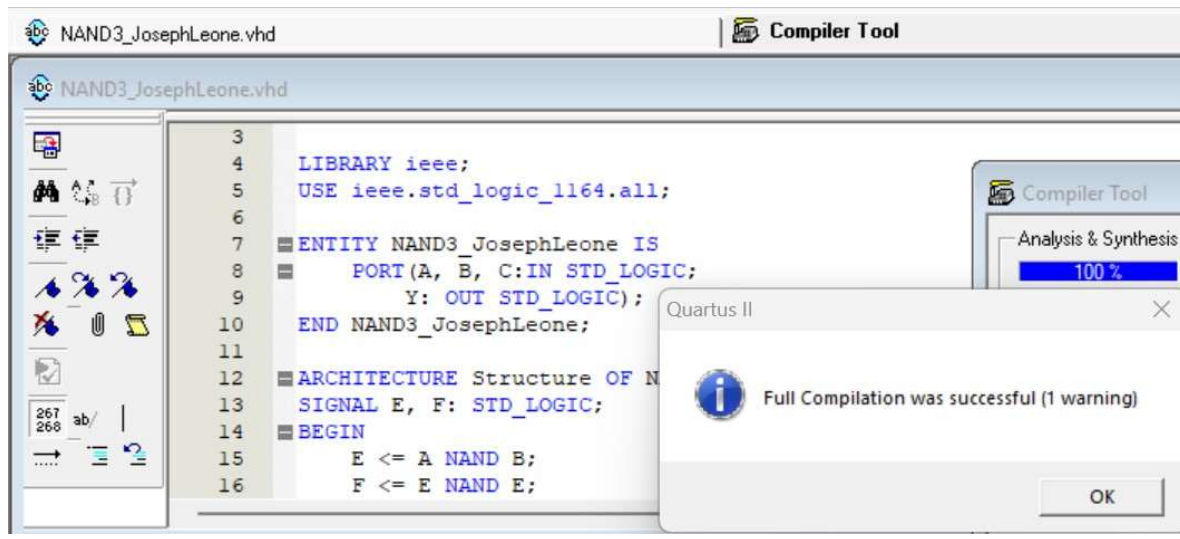
Truth Table (HANDDRAWN)(1pts)

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

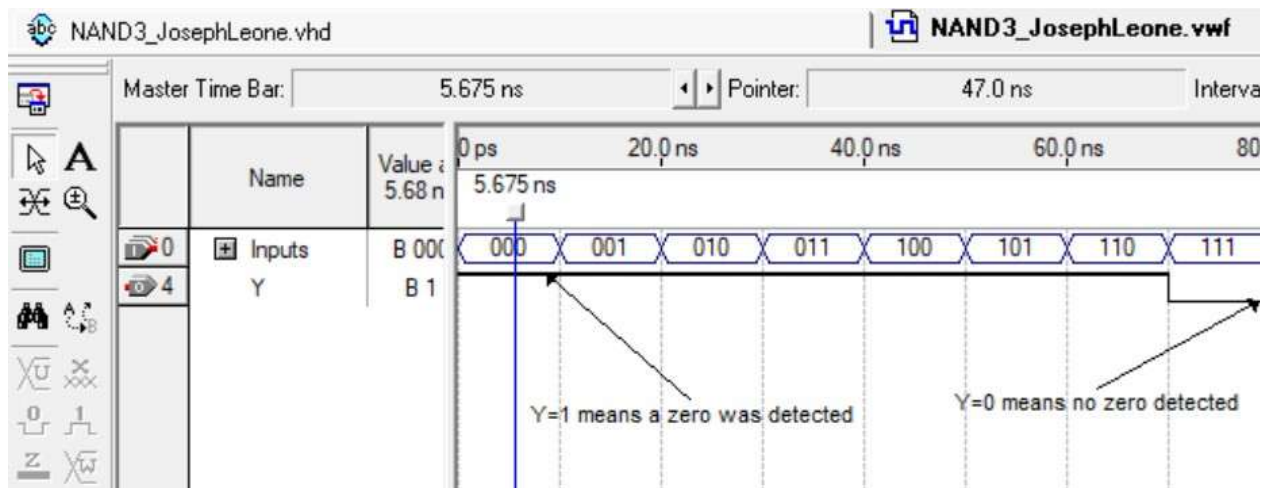
VHDL CODE(2pts)



Schematics Successful Compilation(1pts)



Timing Diagram (vwf)(2pts)



Part 2.2(8pts)-Create a new project in Altera Quartus in your working directory **Lab1A_YourName** using VHDL to define a 4-input NAND circuit called **NAND4_YourName**, using all-NAND gates (**only 2-input NAND gates**).

Project Settings(1pts)

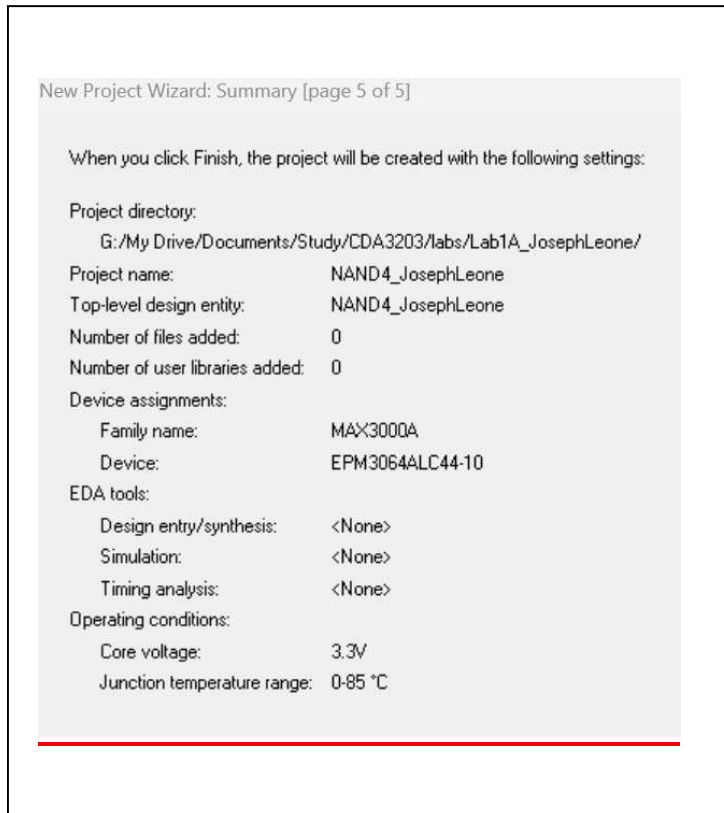
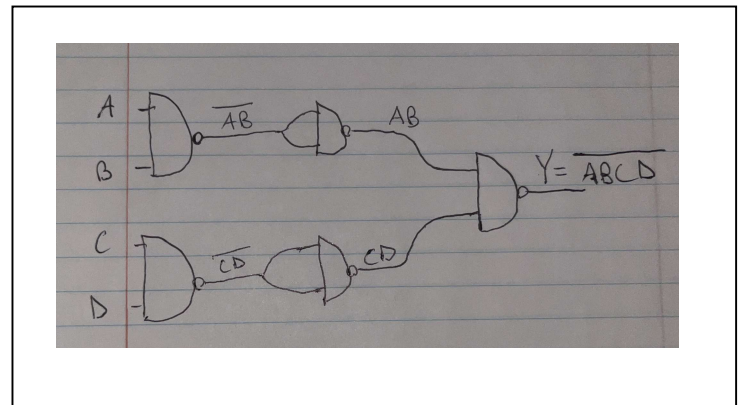


Diagram (HANDDRAWN)(1pts)



Truth Table (HANDDRAWN)(1pts)

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

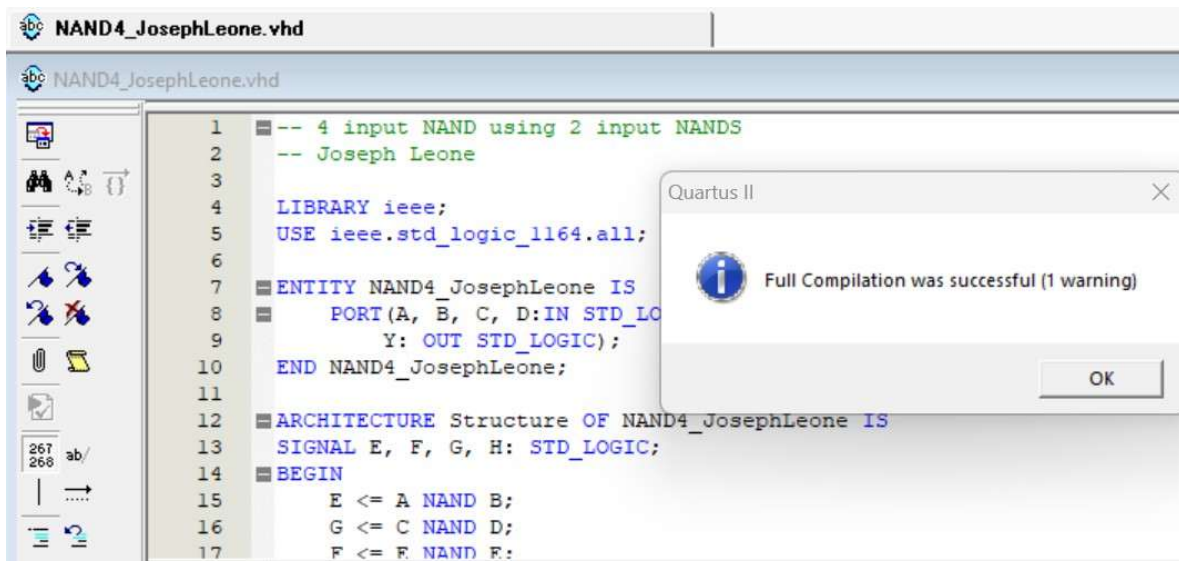
VHDL CODE(2pts)

```

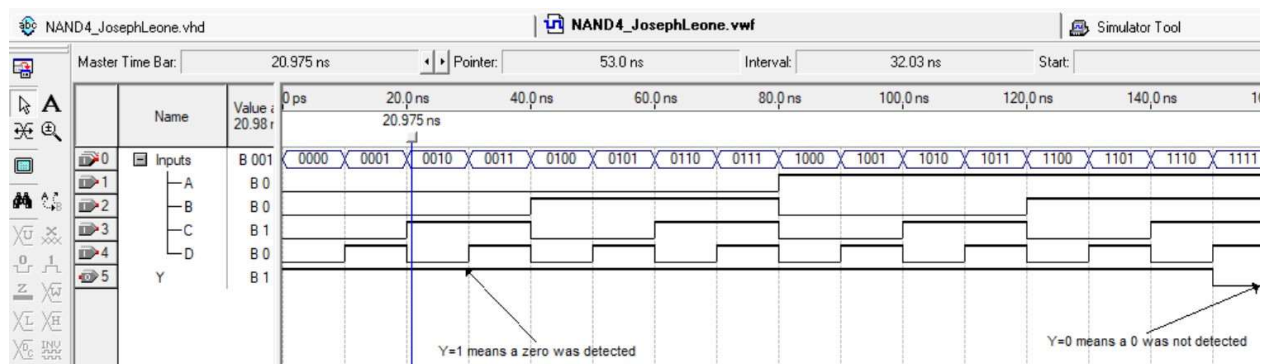
1  -- 4 input NAND using 2 input NANDS
2  -- Joseph Leone
3
4  LIBRARY ieee;
5  USE ieee.std_logic_1164.all;
6
7  ENTITY NAND4_JosephLeone IS
8  PORT (A, B, C, D: IN STD_LOGIC;
9        Y: OUT STD_LOGIC);
10 END NAND4_JosephLeone;
11
12 ARCHITECTURE Structure OF NAND4_JosephLeone IS
13 SIGNAL E, F, G, H: STD_LOGIC;
14 BEGIN
15     E <= A NAND B;
16     G <= C NAND D;
17     F <= E NAND E;
18     H <= G NAND G;
19     Y <= F NAND H;
20 END Structure;
21

```


Schematics Successful Compilation(1pts)



Timing Diagram (vwf)(2pts)



Part 2.3(9pts)-Create a new project in Altera Quartus in your working directory **Lab1A_YourName** using VHDL to define a 2-to-1 Multiplexer/Encoder (Mux) circuit called **Mux2to1_YourName**, using all-NAND gates (**only 2-input NAND gates**).

Project Settings(1pts)

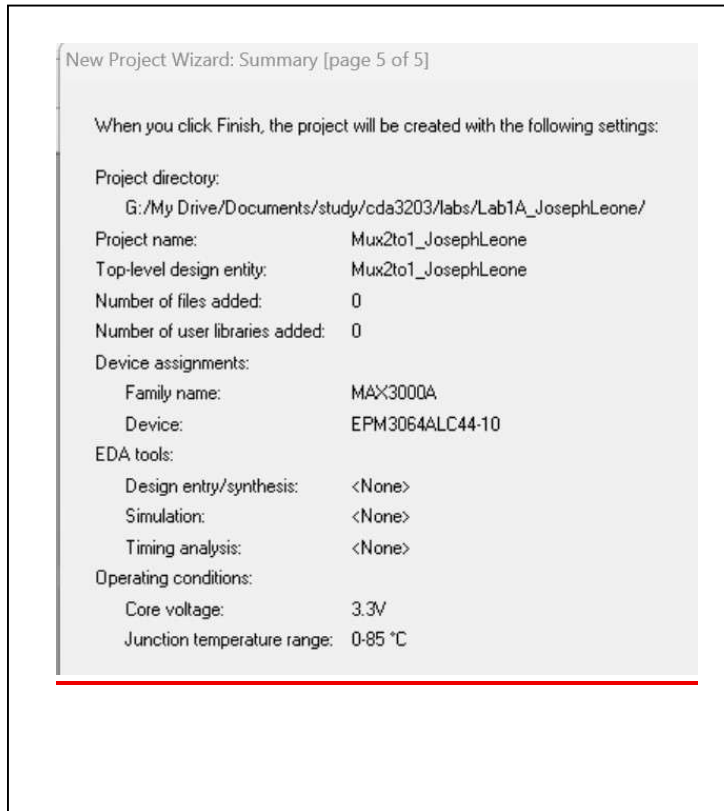
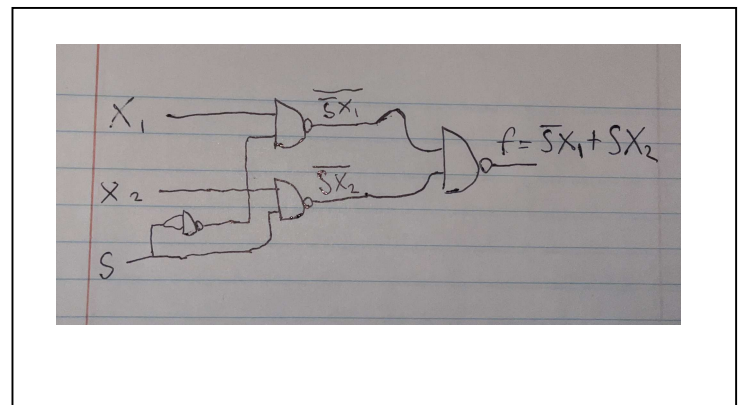


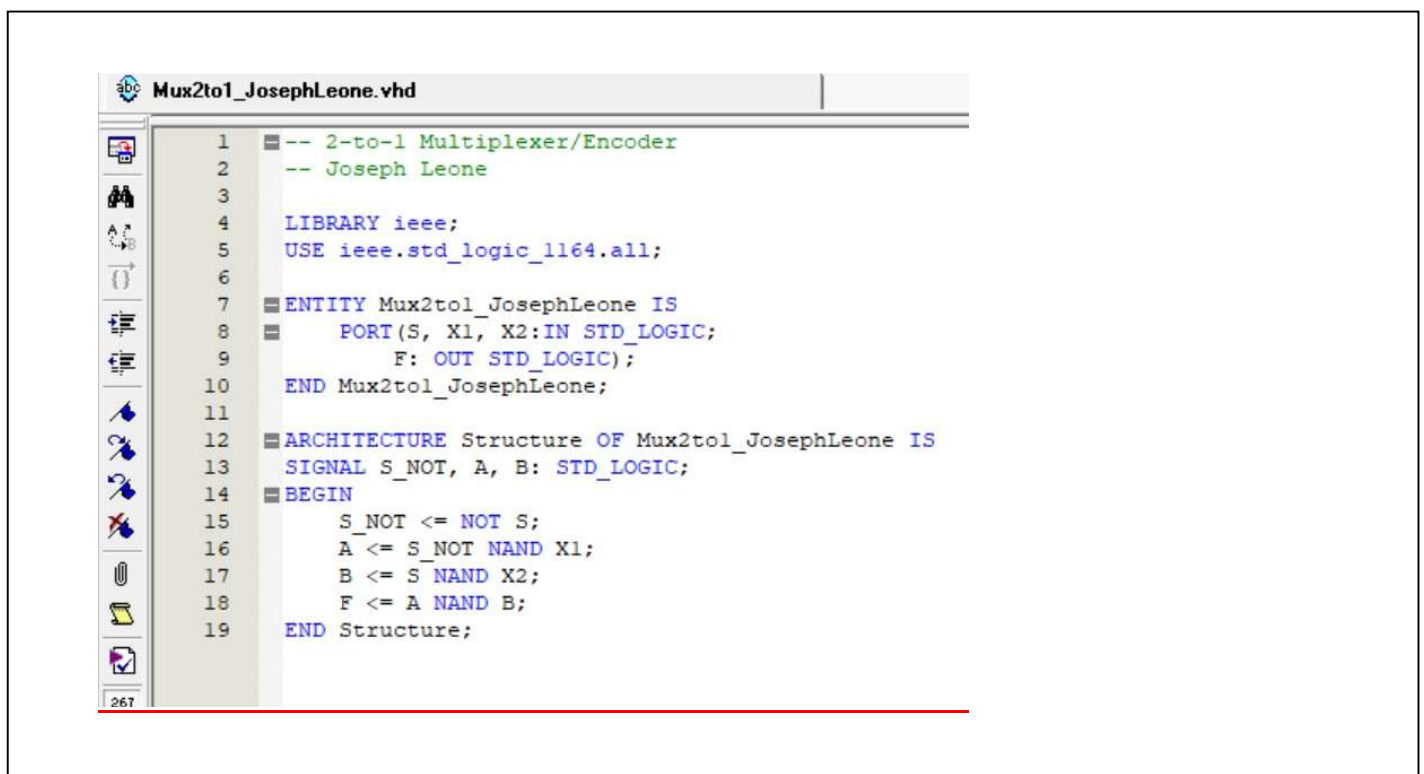
Diagram (HANDDRAWN)(1pts)



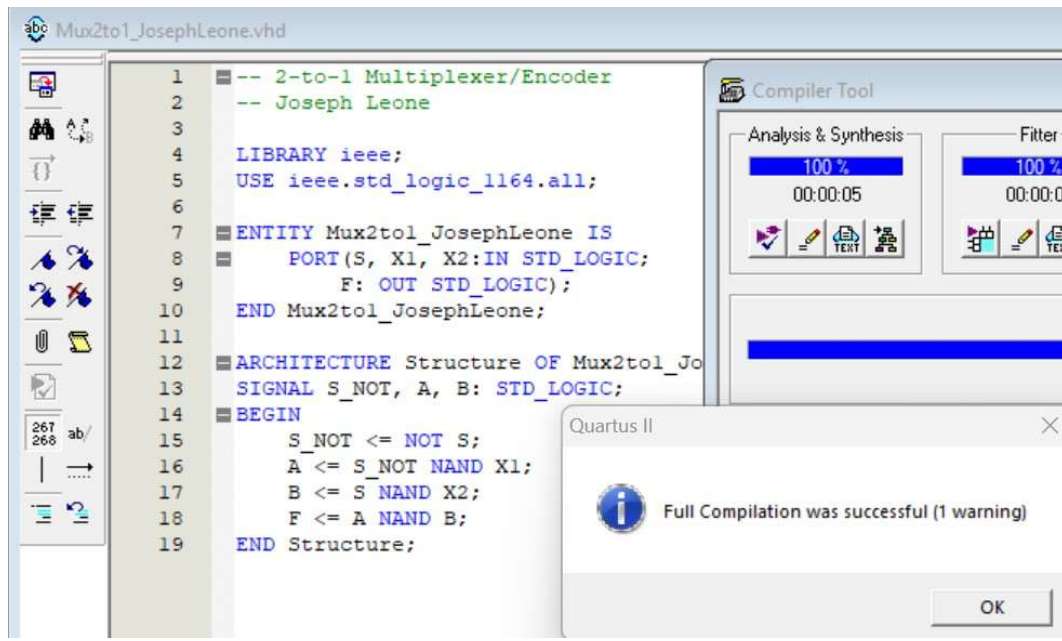
Truth Table (HANDDRAWN)(1pts)

S	X ₁	X ₂	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

VHDL CODE(2pts)



Schematics Successful Compilation(1pts)



Timing Diagram (vwf)(3pts)

